Examiner only

Answer **all** questions.

1. (a) Define the spring constant, k. [1]

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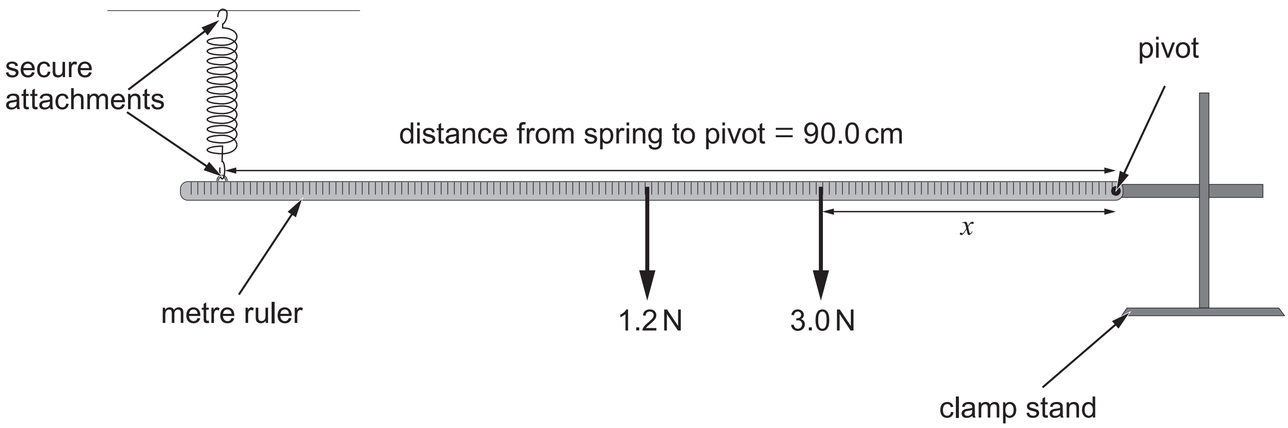
(b) The following apparatus is set up to investigate moments. A horizontal uniform metre ruler weighing 1.2 N is freely pivoted at one end. The ruler is suspended by a spring of spring constant, $k = 20\,\text{N}\,\text{m}^{-1}$, at a point 90.0 cm from the pivot, and a load of 3.0 N is suspended at a distance, x, from the pivot.



Use the following information to calculate x.

Original (unstretched) length of spring = 10.0 cm
Stretched length of spring when the ruler is horizontal = 17.0 cm [4]

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(c) Two springs of the same type used in part (b) are now connected in series as shown.

Laura believes that connecting the springs in this way will result in the overall spring constant being $10\,\text{N}\,\text{m}^{-1}$. Aled believes that the overall spring constant will be $40\,\text{N}\,\text{m}^{-1}$. Explaining your answer determine who, if either, is correct. [2]

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